

Aviation Cybersecurity and Cyber-Resilience: Assessing Risk in Air Traffic Management



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Abstract Civil aviation is the safest transport mode in the world and probably also the most interconnected system of information and communication technology. Cyber-attacks are increasing in quantity and persistence, so the consequences of a successful malicious cyber-attack on civil aviation operations could be severe nowadays. New technologies, extension of connectivity and their integration in the aviation industry, especially in the field of Air Traffic Management (ATM), increase the risk to these critical assets. This chapter examines cyber security challenges and interoperability in ATM systems. We propose an extended threat model for analyzing possible targets and risks involved. We also introduce and analyze cyber resilience aspects in the aviation context and the need for holistic strategy of defense, prevention and response. Under the resilience umbrella, all actors should work on collaborative, risk-based framework to address security threats and increase the aviation systems resilience against future attacks.

Keywords Aviation cybersecurity · Aviation cyber-resilience · ATM cyber threats · Air navigation services cybersecurity

1 Introduction

Security threats to civil aviation operations have become more sophisticated and challenging. One that is emerging in the recent years and arguably even more advanced and complicated to manage is cyber-attack. Today, the global civil aviation community is relying on computer based and information technology (IT) systems

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for their daily frontline and backroom operations. This reliance is expected to grow as new and modern airports are developed, new aircraft introduced into service and stakeholders seek to meet the growing demand of the more IT-savvy passengers with new passenger facilitation processes, using digital and IT-based systems [1].

Aviation is a key foundation for international trade, tourism, and investments crucial to the global economy development. The air transport industry supports 2.7 trillion dollars or 3.5% of the world's gross domestic product (GDP) providing 9.9 million direct jobs within the air transport industry [2]. According to the latest traffic forecasts, by 2034, both air passenger and air freight traffic are expected to double, compared to 2016. Passenger traffic is expected to overpass 14 trillion revenue passenger-kilometres (RPKs) with a growth of 4.5% per annum, and freight will expand by 4.2% annually over the same time period, reaching 466 billion freight tonne-kilometres (FTKs) [2].

The use of Information Technology in civil aviation has also increased exponentially in the last years. Digitalization, technological tools and systems often connected to the internet increase intelligence and interoperability on one hand, while on the other they may constitute serious risks for aviation cyber security. Therefore, it is necessary to keep a high level of attention and awareness on possible future developments of the cyber threat [3].

The overall aim is to reduce the vulnerability to cyber-related risks, to strengthen the air transportation systems resilience against cyber threats, which is seen as the capability of an organizational and technical system to protect itself from failures or losses, to mitigate impacts by adapting to changing conditions and to recover from degradation after the incident [4].

This chapter looks at some of the challenges and concerns about cyber security threats in the aviation sector. While in previous work [5] we have focused our concerns on the ground, analyzing cybersecurity measures and best practices to improve airports cyber resilience, in this research we present advanced services in surveillance systems of Air Traffic Control with the aim to address existing vulnerabilities and dependencies. Our purpose was to introduce and analyze resilience aspects in the aviation sector and then classify already proposed resilience recommendations, based on their technical, organizational, social, and economic dimensions.

The remainder of this article is organized as follows: section “[Understanding ATM interoperability](#)” examines ATM interoperability and recent advances in surveillance systems. Section “[Aviation cyber threat agents](#)” briefly presents related work on aviation cybersecurity and introduces an extended model with cyber-threat agents in the aviation sector. Security measures are presented in section “[Security measures in ATM](#)”, while section “[Cyber resilience in the aviation context](#)” introduces resilience aspects within the aviation context and analyzes existing in literature resilience proposals on several dimensions. Finally, section “[Cyber resilience in the aviation context](#)” concludes resilience analysis and benefits for the aviation sector.

2 Understanding ATM Interoperability

In order for Air Traffic Control to safely manage airspace, each ground located air traffic controller needs to understand the status of each aircraft under their control. Traditionally, Primary and Secondary Surveillance Radar in various layouts have supported air traffic surveillance and management for decades. Both systems were designed at a time when radio transmission required a great financial investment and expertise. Hence, no security thought was given to these legacy systems, since it was presumed that they would remain out of reach. The rise of Software Defined Radio (SDR) voided this assumption and marked the shift from potential attackers being well resourced to those with much less resource and capability [6].

The ongoing move from traditional air traffic control systems, such as radar and voice, towards enhanced surveillance and communications systems using modern data networks, has caused a substantial shift in the security of the aviation environment. Implemented through aviation research programs like the Single European Sky ATM Research (SESAR) and the US American NextGen programs, several new air traffic control and communication protocols are currently being rolled out that have been in the works for decades [7].

In this section, we briefly describe the basic ATM systems serving surveillance and interoperability, used for air traffic control such as: Primary and Secondary Surveillance Radar, Automatic Dependent Surveillance-Broadcast, Traffic Collision and Avoidance System, and Wide Area Multilateration. All these systems interact with each other as graphically presented in Fig. 1. Then we discuss how recent advances in wireless technologies have changed the threat landscape in the aviation context.

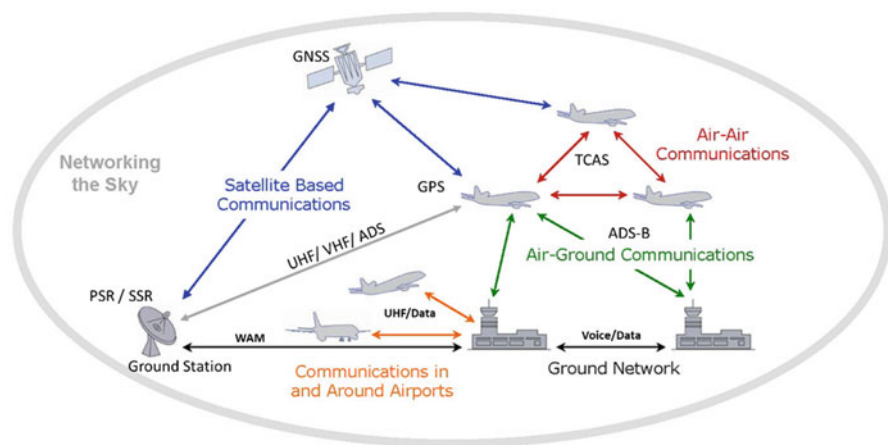


Fig. 1 ATM interoperabilities

Primary Surveillance Radar (PSR) describes non-cooperative radar localization systems. In civil aviation, these typically employ a rotating antenna radiating a pulse position-modulated and highly directional electromagnetic beam on a low GHz band. Potential targets in the airspace reflect the pulses and measurement of the bearing and round-trip time of these reflections provides the target's position. Whilst PSR is not data-rich, it is relatively hard to attack as it relies on physical properties [8].

Secondary Surveillance Radar (SSR) is a cooperative technology with modern communication versions, including the so-called transponder modes. SSR provides more target information on ATC radar screens compared to PSR. Ground stations interrogate aircraft transponders using digital messages on the 1030 MHz frequency, which reply with the desired information on the 1090 MHz channel. Commodity hardware can receive and transmit on these frequencies, making them accessible to attack [9]. Mode S is a particularly important for the current SSR system. It supports systems of increasing significance in modern aviation surveillance, in conjunction with multilateration techniques to provide redundancy. Being intentionally designed with lack of confidentiality, all SSR systems are subject to eavesdropping attacks by passive observers [7].

Automatic Dependent Surveillance-Broadcast (ADS-B) is a protocol in which aircrafts continually broadcast their own ID, position and velocity as well as further information such as intent or urgency codes. These broadcasts do not require interrogation but independently send the aircraft's position and velocity twice a second and unique identification every 5 s; ADS-B is currently in the roll-out phase and it is mandated for use by all aircraft from 2020 in all European and American airspace [7].

Traffic Collision and Avoidance System (TCAS) allows aircraft to interrogate nearby aircraft, in order to resolve airspace conflicts. For example, should another aircraft come within some predefined range, TCAS will initially produce a Traffic Advisory notifying the pilot of traffic nearby. Should the intruder enter the immediate airspace of the aircraft, an alarm will be produced which instructs one of the aircraft to change course. Since the first December 2015, TCAS is mandated for inclusion on civil aircrafts carrying more than 19 passengers or with a minimum take-off weight of 5700 kg [10].

Wide Area Multilateration (WAM) is particularly useful for ATM since it allows location estimation of an aircraft using 1090 MHz messages over large areas. WAM, combined with ADS-B, will form a key part of the next generation surveillance technologies [11] and can help to detect unusual ADS-B reports. Due to the number of sensors and data processing equipment required to cover large areas, the cost of installation is very high, which makes WAM quite hard to attack.

To aggregate information, all the above surveillance systems of Air Traffic Management with discussed characteristics, dependencies and vulnerabilities are presented in Table 1.

Table 1 Main characteristics, dependencies and vulnerabilities of ATM systems

System	Ground/air dependent	Deployment status	Technology	Dependency	Vulnerability
Primary Surveillance Radar (PSR)	Ground	In use	Measure the bearing and distance of targets using the detected reflections of radio signals	Airplane target independent	Not IT related
Secondary Surveillance Radar (SSR)	Ground	In use	Requests additional info from aircraft like identity, altitude, speed	Targets equipped with transponder	Eavesdropping
Traffic Collision and Avoidance System (TCAS)	Air	In use/mandatory since 2015	Target identity interrogation	Targets equipped with transponder	Eavesdropping, jamming, spoofing
Automatic Dependent Surveillance-Broadcast (ADS-B)	Air	Mandate by 2020	Targets broadcast information about identity, altitude, speed	Targets equipped with transponder	Eavesdropping, jamming, spoofing
Wide Area Multilateration (WAM)	Ground	In deployment	Combines ADS with PSR SSR Data for robustness	Central Processing IT based information	Data processing and IT related

3 Aviation Cyber Threat Agents

Although air transportation has a long history of risk management with a special focus on safety and physical security, the field of cyber risks has recently introduced a new landscape of threats. In 2016, at the 39th Assembly, International Civil Aviation Organization (ICAO) has announced preparation works on cybersecurity and cyber resilience. In this direction, Chapter 18 of the Aviation Security Manual which deals with cyber threats has been updated in September 2017. Moreover, Aviation Security Manual (Doc 8973) was enhanced to provide guidance, including minimum measures, to protect critical information systems against unauthorized access and use [11–12].

In addition, recent research studies revealed that cyber threat will most likely be one of the main security issues in aviation, since according to SESAR and NextGen programs the overall air transport system will massively migrate to an IP based infrastructure and operate in accordance with network centric operations concept, with real-time information sharing [4–9]. As a critical resource, information must be treated like any other critical asset which is essential to the efficiency and successful delivery of ATM systems.

In the area of aviation cybersecurity, research work has shown that complexity and criticality of information security and its governance demand the highest organizational security level. Civil Air Navigation Services Organization (CANSO) has issued in 2014 a guidance for Cyber Security [13] explaining how air navigation service providers should take into account cyber security risks in air traffic management, including cyber threats, vulnerabilities, motives of threat actors, as well as considerations about managing cyber risks and implementing a cybersecurity program. Sathish et al. [14] proposed a vulnerability assessment framework for wireless threats in Aviation Cyber-Physical Systems (ACPS) by evaluating the tools and used them in their framework to assess the various threats associated with ACPS. Sampigethaya et al. [15] presented a comprehensive survey of security of the e-enabled airplane with applications such as electronic distribution of loadable software and data, as well as future directions such as wireless networked control and airborne ad hoc networks. Through his approach, Bernard Lim [16] looks at some of the challenges and possible ways to address the concern of cyber security threats confronting the global civil aviation community.

During their study, Stander and Ophoff [17] found that steps are taken by aircraft manufacturers and controlling bodies to prevent the occurrence of incidents as to compromise the information systems of an aircraft. D. Jeyakodi [18] justifies in her work how global aviation industry will remain a target for adversaries seeking to make a statement or cause substantial loss to life and financial bearing. The key to ensuring security would be to keep up with the developments thereby being in a position to confront the threats rather than evoking responsive action after its occurrence.

Strohmeier et al. [7] have presented a realistic threat model based on the up-to-date capabilities of different types of threat agents and their impact on a digitalized

aviation communication system, where threat agents are classified based on their motivation and capabilities. We have extended this model by adding a new threat: “the insider”. We strongly believe that this actor remains a considerable threat agent, not to be neglected from the scheme. We have also estimated risk exposure, taking into account implemented security controls and available security solutions and countermeasures, already proposed in literature. Table 2 presents this extended taxonomy applicable to wireless security in aviation ATM systems and we briefly describe each threat agent characteristics:

- *Passive observers* exploit the open nature of air traffic communication protocols. They use public and private websites and mobile applications, which display air traffic and its communications in real time, to gather information about private or secret air traffic movements. Alternatively, they can employ cheap SDR receivers to gather their own undistorted picture of all air traffic in their vicinity, in real time or stored for later analysis. The information collected can be exploited in many ways, ranging from privacy concerns to the detection of military operations. The risk exposure of ATM systems in such threat agents is rather low, due to no offensive capabilities in the aviation industry.
- *Activists and hobbyists* are the lowest active threat in our model, based on their abilities concerning both hardware and knowledge. Their aim is to exploit security holes with existing, easy-to-use attacks with typically low sophistication and they are able to monitor and interfere to aviation communication channels. Their motivation is regularly not rational, instead any identifiable impact is sought for publicity, thrill and recognition [19]. The risk exposure of ATM systems in such threat agents is considered low, since they can be detected and mitigated with the use of back-up surveillance systems.
- *Insiders* can be a serious threat and are often disgruntled employees, former employees, contractors, or even business associates. These users have inside information of the organization’s security practices, data, and computer systems. Insiders can be greedy, malicious or unpredictable in their motivations. The fact that an insider has access to key applications and other critical systems makes him potentially even more dangerous than third-party cybercriminals who try to break in through malware and other mechanisms. Therefore, risk exposure of ATM systems is medium since it is really hard to promptly detect insider’s malicious intent or actions.
- *Cyber-crime attackers* usually seek to attack systems for monetary gain, having a sufficient knowledge, using software-defined radios, and even small unmanned aerial vehicles (UAV), being able to inject new messages or modify existing ones in such ways that they are not flagged by current detection systems. They try to cause maximum damage and exert credible threats, as a pre-requisite for blackmail or to take advantage of inside knowledge. Consequently, they are seeking to exploit any possible and effective way to attack Air Traffic Control and aircraft systems. The risk exposure of ATM systems is medium and should be seriously taken into consideration in regular performed risk assessments.

Table 2 Threat agents in aviation systems

Threat	Resources	Goal motivation	Capabilities	Hardware cost	ATM target	Risk
Passive observers	Very Low	Information collection financial or personal interest	Eavesdropping, use of website & mobile apps	Internet access, SDR receiver stick (\$10)	ADS-B	Low
Activists and lobbyists	Low	Any noticeable impact thrill and recognition	Eavesdropping, replay attacks, denial of service	COTS SDR transmitter (\$300–\$2,000)	ADS-B	Low
Insiders	Low – Medium	Disgruntlement, revenge, maximise financial gains	Resources for specific impact on operations, based on proprietary knowledge	Low cost, enforced by inside use of tools and info on security gaps	SSR, RSR, ADS-B, TCAS	Medium
Cyber crime	Medium – High	Maximising impact financial gains using e.g. blackmail or valuable information	Resources for large-scale operations with sophisticated transponders	Directional antennas, small UAV's with SDR transmitters (\$5,000)	SSR, PSR, ADS-B, TCAS	Medium
Cyber terrorism	Low – Medium	Political or religious motivation; Massive disruption and casualties	Resources for specific high-impact ops, though usually on a limited scale	As with cyber crime, potentially on a smaller, more targeted scale	SSR, PSR, ADS-B, TCAS	High
Nation state	Unlimited	Weapons targeting specific, potentially military objects	Anything physically and computationally possible	Military-grade radio equipment, capability for electronic warfare	SSR, PSR, ADS-B, TCAS, WAM	High

- *Cyber-terrorists* seek to threaten national security, cause mass casualties, weaken the economy, and damage public morale and confidence in aviation systems [19]. By exploiting the vulnerabilities in wireless aviation communications, terrorist groups, which traditionally hijack or crash planes using physical weapons, could mount attacks on planes from the ground and from safe distances. The risk exposure of ATM systems is high, due to the increased capacity of terrorists and extremists nowadays to use of IT and cyber technologies for their illegal purposes. This tendency is on the rise around the world, attributed to political and social instability in Middle East, North Africa and other conflict areas.
- *Nation state actors* can be part of the electronic warfare threat model, although traditionally this is outside the scope of securing civil aviation [7]. With sufficient knowledge of intrusion detection systems and near-unlimited resources, it is possible to bypass plausibility checks and redundancy-based defenses even in the ATM sector. The risk exposure of ATM systems is considered high and depends rather on specific political circumstances.

4 Security Measures in ATM

Since ATM Security is major component of Aviation Security, it plays a key role in the prevention and response to threats aimed at all parts of the aviation system including national and international high-value assets. In addition, ATM Security has an interface with Airspace Security revolving around national security and defense requirements, providing technological security and interoperability between civil and military systems [7]. Security threats may be directed at aircraft or through them to targets on the ground. The international dimension imposes the uniform and effective application of suitable measures. ATM has to support national security in respect of the identification of flights entering a State's national territory and Air Defense organizations have to be provided with all ATM information relevant to their task [3].

In general, security measures in aviation range across a number of security disciplines. It does not matter if the asset to protect is an aircraft, an airport, a control center or an information network, all security elements apply at a certain degree, as already referenced in Security Standards (ISO, NIST, ISA) and literature recommended practices [12–19]. In Table 3 we brief these basic security measures and disciplines.

While the above security principles can be implemented to a certain efficiency degree, there is a need for a 'holistic view' covering all challenges of aviation security for all phases of air transport, both on the ground and in the air, since the weakest link in the chain is the one likely to break.

Especially the last security element for operational continuity aims to handle degradations of the ATM system. Although it may encapsulate more managerial aspects, is an essential part of the overall aviation security cycle. It highlights the need for a holistic strategy of defense, prevention and response and introduces the

Table 3 Basic security measures and disciplines in ATM systems

Security discipline	Security measures
Physical security	Access control, perimeter protection, screening, control checks, assets responsibility, redundancies, environmental protection
Personnel security	User access management, security clearances, segregation of duties, recruitment policy, staff regulations, vetting, staff awareness and training
Information security	Protection of information CIA: Confidentiality, Availability, Integrity; cryptography, media handling, backups, software updates and patches
Communication security	Network segregation, security management, intrusion detection management, event logging, teleworking and mobile devices policies
Intelligence support	Security without intelligence is meaningless; intelligence support is a transverse requirement for threat assessments, threat watch and security alert levels declaration
Security information exchange	Information exchange between national authorities, security and intelligence organizations and ATM security managers, security warnings, threat and alert levels, incident identification and notification, reporting and incident resolution follow-up
Operational continuity	Emergency response, business continuity management and contingency plans

need for resilience management. The idea of resilience and its related aspects is introduced and analyzed in the next chapter section.

5 Cyber Resilience in the Aviation Context

The idea of cyber-resilience in ICT, in its most basic form, is the evaluation of what happens before, during and after a digitally networked system encounters a threat. Resilience is not event-specific: it accrues over the long term and should be included in overall business or organizational strategy. The different understandings of resilience are described in IMPROVER¹ project taking into account a combination of different properties [20–21]. Some definitions target on foresight, robustness, resourcefulness, redundancy, rapid recovery and adaptability. Others take prevention, preparedness, respond and recovery into consideration. According to IMPROVER, Resilience concepts encompass several dimensions, such as technical, organizational, social, and economic ones, as presented below:

- The *technological* dimension refers primarily to the physical properties of infrastructure components and systems and refers to the characteristics and behavior of these in the case of a change or incident.

¹IMPROVER is a Horizon 2020 project focusing on how to improve European critical infrastructure resilience

- The *organizational* dimension, as it relates to the organizations and institutions that manage the physical components of the systems, i.e. CI operators or owners. It covers aspects such as culture, people, business continuity, risk and disaster management at the organizational level.
- The *social* dimension encompasses population and community characteristics that render social groups either more vulnerable or more adaptable to hazards and disasters.
- The *economic* dimension focus on reducing both direct and indirect economic losses resulting from disasters, in various levels.

In aviation context, Eurocontrol Research program uses for resilience the following definition: “*Resilience* is the ability to prevent disruptions, to prepare for and adapt to changing conditions and to respond and recover rapidly from disruptions to ensure the continuity of services at an acceptable performance level”. The aim of this definition is to achieve the understanding that caring for resilience is more affiliated to the management of risks rather than to the elimination of them [22].

Being resilient implies minimizing reductions in performance (acceptable drop of performance) in the face of a successful attack. This means to be able to work properly also in several levels of degraded mode, while healing measures and repair works can be undertaken. It is therefore essential to provide methods and means to allow the solution to recover, as quick as possible from such degraded modes, achieving minimum recovery time.

As presented in Fig. 2, under the resilience umbrella, the whole set of measures, which are required for sufficient resilience against cyber-attacks, is a combination of different actions and proper behavior before, during and after the incident

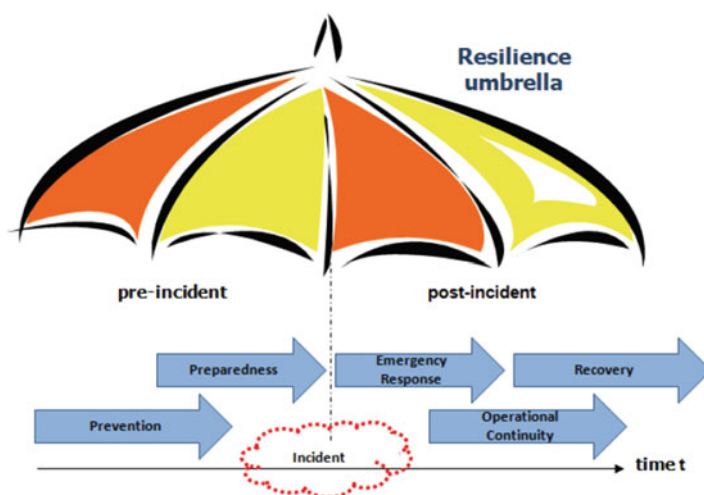


Fig. 2 The resilience umbrella. (Source: EUROCONTROL)

[23]. The flow of cyber resilience actions already starts when the services, tools or systems under concern are in the development phase, which is related to the “security by design”. When taking this into account, the first step of “*Prevention*” is profound established. Controls may have to be put in place to address potential risks emanating from other parts of the system or “system of systems”. Another pillar of resilience is “*Preparedness*” for possible attacks, which can be achieved by procedures and training of staff. Being prepared for any cyber-attack begins with thinking about daily activities and the way work is organized and conducted. This includes also the knowledge about the fastest and most secure ways of decoupling software tools from the system or network and safely/securely shutting down infected systems [23].

When being under attack the “*Emergency Response*” to the attack is also important. The first response focusses on identifying the problem, containing it, eradicating it. Responsive measures may also include the restriction of services or the unwinding of trained sequences. The focus shall be kept on the secure delivery of services and data whilst being aware of the attack in progress. This supports and enhances “*Operational Continuity*”. The response phase needs to be continued until the cause and even the cascading effects of the attack have been eliminated, accounted for or phased out. When at any point in time this can surely be confirmed the phase of “*Recovery*” may be initiated. This phase again needs to be as short as possible in order to have all services, tools and systems in full operation after a cyber-attack.

Although resilience engineering has been introduced in the aviation mainly for enhancing safety sector, it has not been thoroughly expanded to the cybersecurity aspects and resilience in the air traffic management area. Resilience in aviation sector has been partially discussed in previous research [4, 5, 23, 24]. However, in our work we have studied a recent research, the ARIEL project for Air Traffic Resilience, which aimed to perform a holistic risk analysis and evaluation of critical infrastructures in aviation [4]. According to ARIEL, resilience is seen as the ability of a system to absorb or avoid damage without suffering complete failure and integrates the aspects of protection, mitigation and recovery. It proposes a continuous dynamic and model-based cyber risk analysis process, in order to establish persisting capabilities of cyber resilience in the air transportation system. Project report identified the following recommendations, as essential for resilience implementation in the aviation sector, which are listed and briefly explained below:

R-1) Develop the structural and procedural basis for continuous intra- and inter-organizational cyber resilience analysis: Combining classical information security and newly developed cyber operational resilience approaches. Establishing an organizational structure that brings together the personnel of all relevant disciplines inside and across air traffic organizations to cope with the evolving cyber threat landscape in a holistic way. This has to be combined with suitable continuous processes aligned with the existing information security norms and standards.

- R-2) *Develop and manage interdisciplinary cyber risk analysis teams:*** To facilitate the establishment of interdisciplinary collaborating teams, there is a need to develop and apply the necessary methods and management approaches comprising elements of a common language; knowledge management and transfer; ignorance management for balanced evaluation of findings; widespread basic IT knowledge and security awareness by personnel of all disciplines including middle and top management.
- R-3) *Develop and maintain a portfolio of cyber threat scenarios:*** In contrast to the currently applied ad hoc way of threat scenario development and utilization, the introduction of a structured continuous process for the development and evolution of air-traffic cyber threat scenarios is recommended. This is to be combined with suitable methodology to develop scenarios and to apply them in the areas of knowledge development, training as well as verification and validation.
- R-4) *Ensure the interoperability of cyber-relevant models and data:*** Developing standardized meta-models for computer-based data exchange and collaboration of different models is needed. The integration and comparison of cyber-relevant results and findings in tool-based analysis and decision support is also essential. To enable interdisciplinary or even inter-domain collaboration based on a comprehensive approach, data sharing concepts are needed for a reuse of existing data, which include technical, methodological and organizational aspects.
- R-5) *Refine and Evolve Dynamic Risk Analysis Methods:*** Additional effort into the further evaluation and evolution of the model-based dynamic risk analysis method should be developed. This semi-automated analysis method enables to dynamically model and analyze cyber risks in complex systems, large organizations or even in between several interconnected organizations. The high potential of this approach enables a comprehensive, dynamic cyber risk assessment in the aviation sector.
- R-6) *Safety & Security – Ensure consistency and enable synergies:*** Since cyber threats and potential cyber-attacks can have a direct impact on safety-critical system functions. Therefore, the development of a comprehensive risk management approach aligning the formerly separated considerations of safety and security under a common roof is requested.
- R-7) *Enhance design methodologies to ensure resilient system characteristics throughout a complete lifecycle:*** The restructuring of architectures of sociotechnical systems could support cyber resilience in addition to protective measures. Existing approaches of resilience engineering, which focus mainly on human factors in complex systems, have to be extended in a technical sense towards integration of cyber resilience capabilities. Some of the more important aspects to be considered are: the preparation of architectures for ongoing changes; the consideration of mitigation and recovery strategies in the system design; and the addition of system functions supporting the detection of cyber-attacks.
- R-8) *Exploit simulation methodologies to support cyber threat and risk analysis of complex systems:*** To achieve a holistic understanding of the effects of potential cyber-attacks in complex systems, simulation is a valuable method

Resilience Dimension	Technological	Organizational	Social	Economic
ARIEL Resilience Recommendation				
R – 1		✓	✓	
R – 2		✓	✓	
R – 3	✓	✓		
R – 4	✓	✓		
R – 5	✓	✓		
R – 6	✓	✓		
R – 7	✓	✓	✓	
R – 8	✓	✓	✓	✓

Fig. 3 Resilience dimension in ARIEL recommendations

to complement more traditional analysis methods. A widespread application of simulation models for processes and systems should be identified by cyber threat and risk analysis to be critical for system operation. Simulation increases the understanding of the impact of identified cyber threats and supports the validation of risk analysis results. Besides using existing simulation models as standalone tools, it is important to develop simulation models “from gate-to-gate” to support holistic analysis of aviation processes. Finally, using human-in-the-loop simulation is vital with operational staff to research the fundamentals of human factors in the face of potential cyber-attacks.

Based on the ARIEL recommendations, we have analyzed the resilience dimensions which are encompassing, according to IMPROVER Resilience concepts (technical, organizational, social, and economic) and the results of this analysis are presented in Fig. 3.

What we can comprehend from the above table analysis is that most recommendations cover at least two resilience dimensions with the technological and organizational ones to be the most common used. There is a core difference of resilience recommendations from cybersecurity disciplines, which usually handle a single dimension at a time. Resilience measures appear to be a synthesis of interactions, collaboration and evolution of current cybersecurity approaches.

The organizational dimension is common to all recommendations, since cyber resilience is really a matter of effective risk management, combined with collaborative working and interdisciplinary strategies to ensure contingency and efficient business continuity.

While there is a lack of recommendations that enforce the economic dimension of aviation resilience, the social dimension is also less developed. The only recommendation that covers all four resilience dimensions is the last one, about achieving a holistic understanding of the effects of potential cyber-attacks with

simulation methodologies using human in the loop, which better support cyber risk analysis of complex systems.

For promoting an overall cyber-resilience approach in the aviation sector, long-term strategy should combine all resilience dimensions that is technological, organizational, societal, and economic. This cyber-resilience approach can ensure greater performance and readiness, making systems more efficient and effective.

6 Conclusions

Recent approaches to increase capacity and efficiency of the existing air traffic system have led to an enormous effort of transition towards digitalization and automation. As a result, formerly separated IT systems get connected via newly established networks for information and data exchange. Due to a growth of complexity, the attack surface of the overall aviation system is increasing and previously unknown interdependencies are being created. Limiting security risk management to “traditional” physical aspects like air terrorism is no longer sufficient to ensure a stable and robust operation of the air transportation system. The component of cyber-security has to be expanded from traditional risk mitigation approaches to more resilient focused approaches.

The domains of air transportation and cybersecurity are organized with a strong focus on protective mechanisms, in terms of their operational and technical implementation. To fulfil the requirements of continuous adaption to a rapidly changing threat environment, the architectures of operational and technical systems have to be restructured based on the results and dynamic simulation risk analysis. According to that, we strongly recommend to balance the cost and performance-driven development and prioritize a sustainable, comprehensive and continuous improvement in order to improve the overall system’s cyber resilience.

As both safety and security are drivers for the determination of resilience requirements, it is sensible to take an integrated view on both subjects to foster the consistency of resilience concepts in aviation.

Since cyber resilience is really a matter of risk management, there isn’t a single point at which it begins or ends. Instead, it comes from building strategy and working to ensure that the risk-transfer mechanisms that work for more traditional threats are also brought to bear on new cyber threats.

Being resilient requires those at the highest levels of a company, organization or government to recognize the importance of avoiding and mitigating risks. While it is everyone’s responsibility to cooperate, in order to ensure greater cyber resilience, leaders who set the strategy for an organization are ultimately responsible and have increasingly been held accountable for including cyber resilience in organizational strategy.

References

1. Lim B (2014) Aviation security – emerging threats from cyber security in aviation – challenges and mitigations. *J Aviat Manag*
2. Industry High-Level Group (IHLG) (2017) Aviation benefits 2017 report
3. De Zan T, d'Amore F, Di Camillo F (2015) The Defence of civilian air traffic systems from cyber threats
4. Kreuzer M, Kiesling T (2017) Recommendations to strengthen the cyber resilience of the air traffic system, ARIEL, Air Traffic Resilience
5. Lykou G, Anagnostopoulou A, Gritzalis D (2018) Implementing cyber-security measures in airports to improve cyber-resilience, WIIoTS in the 2nd global IoT summit
6. Strohmeier M et al (2014) Realities and challenges of nextgen air traffic management: the case of ADS-B. *IEEE Communications Magazine* 52(5):111–118
7. Strohmeier M et al (2016) Assessing the impact of aviation security on cyber power. In the 8th international conference on cyber conflict cyber power
8. Strohmeier M et al (2016) On perception and reality in Wireless air traffic communications security
9. Costin A, Francillon A (2012) Ghost is in the air(traffic): on insecurity of ADS-B protocol and practical attacks on ADS-B devices. In black hat USA
10. The European Commission (2011) Commission regulation laying down common airspace usage requirements and operating procedures for airborne collision avoidance, no. 1332. European Union
11. International Civil Aviation Organisation (2013) Initial capability for ground surveillance. In global air navigation plan 2013–20282
12. International Civil Aviation Organization (ICAO) (2017) Aviation security manual, 10th edition, <https://www.icao.int/Security/SFP/Pages/SecurityManual.aspx>
13. CANSO (2014) Cyber security and risk assessment. Civil Air Navigation Services Organization
14. Kumar S, Xu B (2017) Vulnerability assessment for security in aviation cyber-physical systems. *IEEE 4th international conference on cyber security and cloud computing*
15. Sampigethaya K, Poovendran R, Bushnell L (2008) Secure operation, control and maintenance of future E-enabled airplanes, Network Security Lab (NSL), EE Department, University of Washington, Seattle
16. Lim B (2014) Aviation security – emerging threats from cyber security in aviation – challenges and mitigations, *J Aviat Manag*
17. Stander A, Ophoff J (2016) Cyber security in civil aviation
18. Jeyakodi D (2015) Cyber security in civil aviation
19. Stouffer K, Falco J, Scarfone K (2007) Guide to industrial control systems (ICS) security. *Recomm. NIST.*, no. SP 800–82, pp 1–157
20. Theocharidou M et al (2016) D1.3-final lexicon of definitions related to critical infrastructure resilience, IMPROVER, European Union's horizon 2020 research
21. Lange D et al (2017) Framework for implementation of resilience concepts to critical infrastructure, IMPROVER, European Union's horizon 2020 research
22. EUROCONTROL (2012) Manual for national ATM security oversight, Eurocontrol Publications
23. EUROCONTROL (2009) A white paper on resilience engineering for ATM. <https://www.eurocontrol.int/sites/default/files/article/content/documents/nm/safety/safety-a-white-paper-resilience-engineering-for-atm.pdf>
24. Delgado L, Cook A, Tanner G, Cristóbal S (2016) Quantifying resilience in ATM, in the proc. of 6th SESAR innovation days, Technical University of Delft, The Netherlands